ModelIt!

Teacher's Guide

Why Do Athletes Collapse in the Heat?

Modeling Homeostasis and the Body's Breaking Point

Quick Reference

Grade Level	Grade 09L1
NGSS Standard	HS-LS1-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-3

Plan and conduct an investigation to provide evidence that feedback mechanisms maintain homeostasis.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students plan investigations using their model to test how different combinations of exertion and hydration affect thermoregulation outcomes.
Disciplinary Core Idea	**LS1.A Structure and Function** — Feedback mechanisms maintain the body's internal conditions within a range that supports life, including thermoregulation through sweating and blood vessel dilation.
Crosscutting Concept	**Stability and Change** — Students model how negative feedback maintains homeostatic stability and identify the conditions that push the system past its tipping point.

CAST Testing Alignment

MULTIPLE CHOICE:

An athlete's core temperature rises during a game but returns to normal during halftime. Which feedback mechanism best explains this observation?

CONSTRUCTED RESPONSE:

Using your model, explain why an athlete who is dehydrated is more likely to experience heat stroke than one who is well-hydrated, even if they are performing at the same intensity level.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The human body is a remarkable self-regulating machine. Every second, billions of biochemical reactions generate heat as a byproduct. The thermoregulatory system must continuously dissipate this heat to keep core temperature within a narrow safe range — typically 36.5–37.5°C. When this system fails, the consequences can be life-threatening.

Thermoregulation Basics

The hypothalamus acts as the body's thermostat, detecting blood temperature and triggering cooling responses. When core temperature rises, the hypothalamus signals sweat glands to produce sweat and blood vessels near the skin to dilate, increasing heat loss through evaporation and radiation. This is a classic negative feedback loop.

The Role of Hydration

Sweat is about 99% water. An athlete can lose 1-2 liters of sweat per hour during intense exercise. When fluid loss exceeds intake, blood volume drops, making it harder for the cardiovascular system to transport heat from muscles to the skin surface. The body's cooling capacity diminishes rapidly with dehydration.

Heat-Related Illness Spectrum

Heat illness exists on a continuum: heat cramps (muscle spasms from electrolyte loss), heat exhaustion (heavy sweating, weakness, nausea — body still attempting to cool), and heat stroke (core temp above 40°C, sweating stops, organ damage begins). Heat stroke is a medical emergency with a mortality rate of 10-50% if untreated.

Individual Variation

Not all athletes respond to heat the same way. Factors include body composition, cardiovascular fitness, heat acclimatization (which takes 10-14 days of gradual exposure), hydration habits, medication use, sleep quality, and genetics. This explains why one player collapses while teammates seem fine.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify core temperature, sweat rate, hydration level, and physical exertion as the key components of the body's thermoregulation system.

E **Establish Relationships**
Students determine that physical exertion increases core temperature, which increases sweat rate, which decreases hydration level — and that low hydration breaks the cooling loop.

V **Visualize & Model**
Students build a computational model showing the negative feedback loop of thermoregulation and the conditions under which it fails.

E **Evaluate Outcomes**
Students run moderate/hydrated, intense/dehydrated, and gradual dehydration scenarios to test their model and identify the tipping point.

R **Revise & Extend**
Students add ambient temperature, blood volume, or acclimatization level to explore more realistic thermoregulation dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"In 2022, over 9,000 high school athletes were treated for heat-related illness. Some of them died. Why?"*

Watch For

Show a brief news clip or headline about a student athlete heat collapse. Let the room react.

Slide 2: Learning Objectives

3 min

Say: *"Today you're modeling something your body does every second of every day — regulate its own temperature."*

Watch For

Have students read objectives. Pre-teach 'homeostasis' and 'feedback mechanism.'

Slide 3: Big Question

3 min

Say: *"Same team, same field, same temperature. So why does one player go down while another is fine?"*

Watch For

Quick-write: Students list 3 factors they think determine who collapses. Share out.

Slide 4: LEVER Framework

2 min

Say: *"We're going to build a model that can actually predict when an athlete is in danger."*

Watch For

Preview each LEVER step. Emphasize that feedback loops are the key to understanding this.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key players in the body's cooling system? Which ones can you control?"*

Watch For

Guide sorting of external vs. internal components. Discuss why hydration is external (you can control it).

Slide 6: Activity 2: Connections

8 min

Say: *"When you sprint in 100-degree heat, what chain reaction happens inside your body?"*

Watch For

Help students trace the feedback loop. Identify where the loop breaks.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's find the exact point where the body's cooling system fails."*

Watch For

Students predict tipping points first, then run simulations. Compare results.

Slide 8: Discoveries

5 min

Say: *"So what's the difference between heat exhaustion and heat stroke? Your model just showed you."*

Watch For

Lead discussion connecting model findings to real athletic scenarios. Discuss prevention.

Slide 9: STEM Challenge

12 min

Say: *"Your school just hired you to make sure no athlete ever collapses on this field again."*

Watch For

Groups design monitoring systems using model insights. Present prototypes.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Hydration Level, Physical Exertion
- **INTERNAL:** Core Temperature, Sweat Rate

Relationships

- **Physical Exertion to Core Temperature:** POSITIVE (+) — Higher physical exertion generates more metabolic heat, directly raising core temperature. Muscle contractions convert chemical energy to kinetic energy and heat.
- **Core Temperature to Sweat Rate:** POSITIVE (+) — As core temperature rises, the hypothalamus triggers increased sweat production to cool the body through evaporative cooling.
- **Sweat Rate to Hydration Level:** NEGATIVE (-) — Higher sweat rate depletes the body's water reserves faster, decreasing hydration level. Without fluid replacement, the body loses its ability to cool itself.

Key Discoveries

- The body's thermoregulation is a negative feedback loop — rising temperature triggers sweating, which cools the body back down
- Dehydration breaks the feedback loop because the body can't produce sweat without water, causing temperature to spike
- There's a critical tipping point where the cooling system fails — core temperature rises uncontrollably, leading to heat stroke
- Individual differences in fitness, acclimatization, and hydration habits explain why some athletes collapse while others don't

Multiple Choice Pre-Assessment Answer Key

1. A student claims that sweating is a sign of being out of shape. Which of the following best explains why this claim is scientifically inaccurate?

Correct: A) Sweating is the body's primary mechanism for evaporative cooling and is more efficient in well-conditioned athletes

Correct. Sweating is the body's primary thermoregulatory response, and trained athletes often sweat more efficiently because their cooling systems are better adapted to dissipate heat.

2. During a football game on a hot day, two players of similar fitness levels are performing at the same intensity. Player A collapses from heat exhaustion while Player B continues playing. Which factor most likely explains this difference?

Correct: B) Player A consumed significantly less water before and during the game than Player B

Correct. Hydration level is the most common variable that determines individual vulnerability to heat illness. Dehydration reduces the body's ability to produce sweat and maintain core temperature.

3. Which of the following best describes the concept of homeostasis?

Correct: C) The body's ability to maintain stable internal conditions within a functional range despite changing external conditions

Correct. Homeostasis is the maintenance of stable internal conditions within a functional range, even as external conditions change. It is dynamic, not static, and operates continuously.

4. An athlete's core body temperature rises from 37°C to 39°C during a workout. The body responds by increasing sweat production, which eventually brings the temperature back toward 37°C. This process is an example of which type of biological mechanism?

Correct: B) Negative feedback, because the response counteracts the original change

Correct. This is a negative feedback mechanism because the body's response (increased sweating) opposes the initial change (rising temperature), working to restore the original set point.

5. A coach notices that heat-related illness incidents at their school have occurred at outdoor temperatures as low as 78°F (26°C). Which explanation best accounts for this observation?

Correct: B) Air temperature alone does not determine heat illness risk; humidity, exertion level, and hydration status interact to determine vulnerability

Correct. Heat illness results from the interaction of multiple factors including humidity (which reduces evaporative cooling efficiency), exercise intensity (which generates metabolic heat), and hydration status. Air temperature is only one variable.

Multiple Choice Post-Assessment Answer Key

1. In a computational model of thermoregulation, a student sets Physical Exertion to maximum and Hydration Level to low. The model shows Core Temperature rising sharply while Sweat Rate initially increases then drops to near zero. Which explanation best interprets this model behavior?

Correct: B) The negative feedback loop has broken down because the body cannot produce sweat without adequate water, causing an uncontrolled temperature increase

Correct. This model output demonstrates the critical failure point in thermoregulation. Sweat production requires water. When hydration is depleted, the negative feedback loop breaks because the effector (sweat glands) can no longer function, converting the system to a dangerous positive feedback scenario.

2. A student's model shows that during the 'Gradual Dehydration' scenario, core temperature remains stable for the first 45 minutes, then begins rising rapidly. Which concept best explains this pattern?

Correct: A) The system has a tipping point where the feedback loop transitions from maintaining homeostasis to failing catastrophically

Correct. The model reveals a critical tipping point. Initially, the body has enough hydration to maintain the negative feedback loop. As water depletes, the system reaches a threshold where cooling capacity can no longer match heat production, and the system shifts from stable homeostasis to runaway heating.

3. Two students build models of thermoregulation. Student A's model includes only Core Temperature and Sweat Rate. Student B's model adds Hydration Level and Physical Exertion. Which statement best evaluates the difference between these models?

Correct: B) Student B's model can predict system failure conditions that Student A's model cannot, because it includes the external inputs that drive and constrain the feedback loop

Correct. Student B's model includes the external variables (hydration and exertion) that determine whether the feedback loop functions or breaks down. Without these inputs, Student A's model cannot predict the conditions under which homeostasis fails.

4. Based on model evidence, which intervention would be most effective at preventing heat-related illness during a high-intensity outdoor practice in 90°F weather?

Correct: B) Implementing mandatory hydration breaks every 15 minutes, because maintaining hydration preserves the integrity of the negative feedback loop

Correct. The model demonstrates that hydration is the critical variable that determines whether the thermoregulatory feedback loop continues to function. Mandatory hydration breaks maintain the body's ability to produce sweat and sustain evaporative cooling.

5. A student extends their thermoregulation model by adding an Acclimatization Level component. They find that higher acclimatization shifts the tipping point, allowing athletes to exercise longer before the feedback loop fails. Which reasoning best explains this model prediction?

Correct: B) Acclimatization improves sweating efficiency and plasma volume, allowing the cooling system to operate effectively at higher core temperatures and lower hydration levels

Correct. Heat acclimatization produces physiological adaptations including earlier onset of sweating, increased sweat volume, expanded plasma volume, and improved cardiovascular efficiency. These changes extend the functional range of the feedback loop before it reaches its tipping point.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Sweating means you're out of shape"	Sweating is actually a sign of an efficient cooling system. Well-trained athletes often sweat MORE than untrained people because their bodies have adapted to dissipate heat more effectively. Less sweating during intense exercise can actually be a danger sign.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Drinking water when thirsty is enough"	By the time you feel thirsty, you're already 1-2% dehydrated, which is enough to impair performance and thermoregulation. Thirst is a lagging indicator — the body needs water before the brain signals the sensation of thirst.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Heat stroke only happens in extreme heat"	Heat stroke can occur at surprisingly moderate temperatures when humidity is high, exertion is intense, or the athlete is dehydrated. Multiple cases have occurred at temperatures below 80°F. It's the combination of factors, not just air temperature, that matters.